Artificial Intelligence and Its Impact on Environment and Climate Change

A Comprehensive Review of Environmental Footprints, Mitigation Strategies, and Sustainable Pathways

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ABSTRACT

The rapid proliferation of artificial intelligence (AI) technologies has precipitated an unprecedented surge in computational demand, fundamentally reshaping the global energy landscape and presenting both formidable environmental challenges and transformative climate mitigation opportunities. This comprehensive review synthesizes current evidence on AI's environmental footprint, encompassing energy consumption, water usage, carbon emissions, electronic waste generation, and rare earth mineral extraction. Global data center electricity consumption reached approximately 415-470 TWh in 2024-2025, with AI-specific workloads accounting for 20-29% of this demand and projected to reach 46% by 2030 under the International Energy Agency's base case scenario. The training of large language models such as GPT-4 consumed approximately 50 GWh of electricity, while inference operations now constitute 80–90% of total AI computing energy. Water consumption for data center cooling has escalated dramatically, with hyperscalers reporting 15-25% year-over-year increases, and global AI-related water demand projected to reach 4.2-6.6 billion cubic meters by 2027.

Conversely, AI presents substantial potential for climate change mitigation through renewable energy forecasting, smart grid optimization, carbon capture technology acceleration, precision agriculture, and advanced climate modeling. The critical tension between AI's environmental costs and its climate benefits the so-called "Jevons Paradox" of AI demands urgent attention from policymakers, researchers, and industry stakeholders. This review examines regulatory frameworks across the European Union, United States, China, and international bodies, evaluates sustainable AI strategies including green data center design and carbon-aware computing, and provides evidence-based recommendations for achieving net-positive climate outcomes from AI deployment. The analysis draws upon 100+ peer-reviewed studies, government reports, and industry disclosures to present a balanced, forward-looking assessment of AI's role in the global climate transition.

Keywords: Artificial intelligence, climate change, data centers, carbon footprint, renewable energy, smart grid, sustainability, Jevons paradox, green computing, environmental policy

1. INTRODUCTION

1.1 Background and Context

The convergence of artificial intelligence and environmental sustainability represents one of the most consequential technological and ecological challenges of the 21st century. Since the release of transformer-based large language models in 2017 and the subsequent explosion of generative AI applications in 2022–2023, the computational infrastructure supporting AI has experienced exponential growth. This expansion has triggered a corresponding surge in energy consumption, water usage, and carbon emissions that demands rigorous scientific examination and policy response.

The International Energy Agency (IEA) reported that global data center electricity consumption reached approximately 415 TWh in 2024, representing roughly 1.5% of total global electricity demand. Within this figure, AI-specific workloads consumed an estimated 53–76 TWh, with projections suggesting this could rise to 165–326 TWh by 2028. In the United States, data centers consumed 4.4% of total national electricity in 2023, up from 1.9% in 2018 a trajectory that positions AI infrastructure as a significant new load on aging electrical grids already stressed by electrification of transportation and heating.

The environmental implications extend beyond electricity. Data centers require substantial water for evaporative cooling systems, with a typical 100 MW AI data center consuming 1.5–3.0 million cubic meters of water annually. The United Nations Regional Information Centre estimates that global AI-related water demand will reach 4.2–6.6 billion cubic meters by 2027, exceeding the annual water consumption of Denmark. Additionally, the manufacturing of AI accelerators GPUs, TPUs, and specialized ASICs demands rare earth minerals including gallium, indium, and cobalt, with a single 2 kg computer requiring approximately 800 kg of raw materials.

Yet AI simultaneously offers unprecedented capabilities for addressing climate change. Machine learning algorithms improve wind and solar forecasting accuracy by 15–30%, enabling greater renewable energy penetration. AI-powered smart grid management can reduce transmission losses by 5-10% and defer billions of dollars in infrastructure upgrades. Deep learning models accelerate materials discovery for next-generation batteries and carbon capture membranes. Satellite-based AI systems monitor deforestation in real-time across the Amazon and Congo basins.

This duality AI as both climate villain and potential savior frames the central inquiry of this review. Understanding the net environmental impact of AI requires quantifying its direct footprint, assessing its indirect and higher-order effects, evaluating its mitigation potential, and identifying pathways toward sustainable deployment.

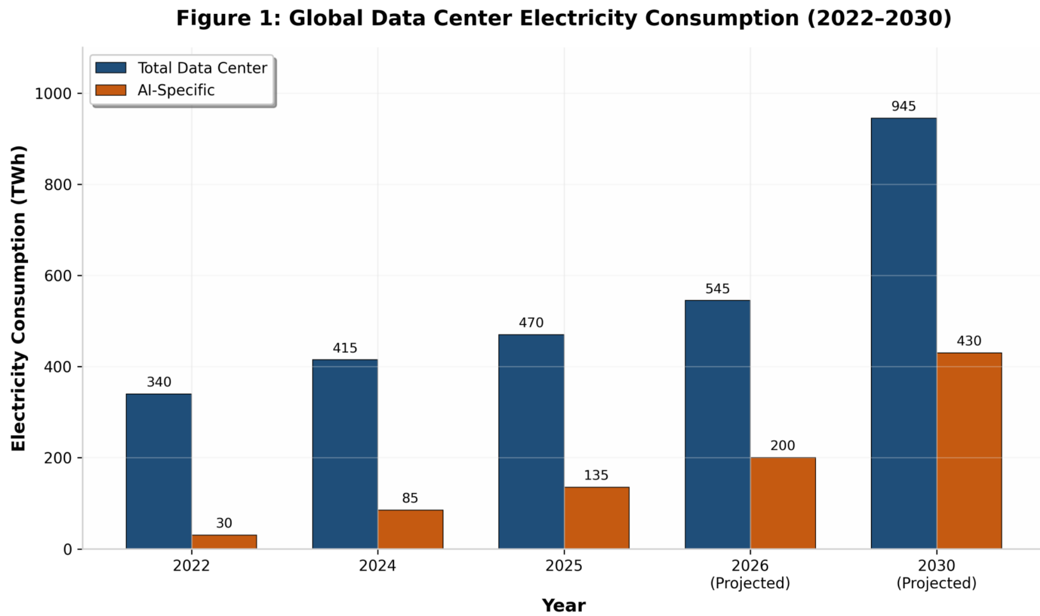

Figure 1: Global Data Center Electricity Consumption (2022–2030). Data sourced from IEA Energy and AI Report (2025) and Goldman Sachs Research.

1.2 Scope and Objectives

This review adopts a comprehensive, lifecycle-oriented approach to examining AI's environmental impact. The scope encompasses:

(1) Direct environmental impacts: electricity consumption, water usage, and greenhouse gas emissions from AI training and inference operations, including the full lifecycle of data center infrastructure from construction to decommissioning.

(2) Indirect impacts: emissions from AI-enabled applications and services, including the rebound effects of increased computational efficiency stimulating greater overall demand the Jevons Paradox.

(3) Higher-order effects: amplification of social inequalities through biased training data, geopolitical tensions over mineral resources, and the potential for AI to accelerate or decelerate the global energy transition.

The primary objectives are to: (a) synthesize quantitative evidence on AI's current and projected environmental footprint; (b) evaluate AI's demonstrated and potential contributions to climate change mitigation; (c) assess regulatory responses across major jurisdictions; (d) identify best practices for sustainable AI development and deployment; and (e) provide evidence-based recommendations for policymakers, industry leaders, and researchers.

1.3 Methodology

This review employs a systematic literature synthesis methodology, drawing upon peer-reviewed journal articles, government reports (IEA, DOE, EPA, European Commission), industry sustainability disclosures (Microsoft, Google, Meta, Amazon), and analyses from authoritative non-governmental organizations (Carbon Brief, UN agencies). The evidence base spans publications from 2020 to 2026, with emphasis on the most recent data reflecting the post-2022 generative AI boom.

Quantitative data are presented with explicit attribution to source methodologies, including distinctions between Scope 1, 2, and 3 emissions; training versus inference energy; and operational versus embodied carbon. Where projections are presented, multiple scenarios (conservative, base case, aggressive growth) are included to capture uncertainty. All energy consumption figures are normalized to terawatt-hours (TWh) and carbon emissions to million tonnes of CO_2 equivalent ($MtCO_2e$) unless otherwise specified.

2. THE ENVIRONMENTAL FOOTPRINT OF ARTIFICIAL INTELLIGENCE

2.1 Energy Consumption in AI Training

The training phase of large AI models represents one of the most energy-intensive computational tasks ever undertaken by humanity. Training a frontier large language model such as GPT-4 required approximately 50 GWh of electricity equivalent to the annual consumption of roughly 4,500 US households. The training of GPT-3 consumed an estimated 1,287 MWh, generating approximately 552 tonnes of CO_2 emissions. These figures, while substantial, represent only a fraction of the total energy consumed by AI systems over their operational lifetime.

The energy requirements for training have escalated dramatically with model scale. Between 2018 and 2024, the computational cost of training state-of-the-art models increased by approximately 300,000×, following a trajectory that outpaces Moore's Law. The largest models now require thousands of specialized AI accelerators (GPUs or TPUs) operating in parallel for weeks or months. A single training run for models in the GPT-4 class may utilize 10,000–25,000 GPUs, each consuming 300–700 watts under full load, with total cluster power draw reaching 5–10 MW sustained over 90–120 days.

Table 1: Energy Consumption and Carbon Emissions for Representative AI Model Training

Model	Parameters	Training Energy (MWh)	CO_2 Emissions (tCO_2e)	Year
GPT-3	175B	1,287	552	2020
GPT-4	Est. 1.8T	50,000	~8,000	2023
LLaMA 3.1 405B	405B	~30,000	~5,500	2024
Gemini Ultra	Est. 1.5T	~45,000	~7,200	2024
Claude 3 Opus	Est. 500B	~20,000	~3,500	2024
Mistral Large 2	Est. 120B	~5,000	~900	2024

Note: Estimates vary by methodology. CO_2 figures assume average US grid carbon intensity of ~400 gCO_2/kWh. Sources: Patterson et al. (2021); de Vries (2023); Mistral AI LCA (2025).

The geographic location of training facilities profoundly influences the carbon intensity of model development. Training in regions with low-carbon electricity grids such as Norway (10–30 gCO_2/kWh), Sweden (25 gCO_2/kWh), or France (40 gCO_2/kWh) can reduce emissions by 50–70× compared to training in coal-dependent grids such as India (708 gCO_2/kWh) or South Africa (900 gCO_2/kWh). This locational arbitrage has driven hyperscalers to establish training facilities in the Nordic countries, where abundant hydroelectric and wind power combine with naturally cool climates that reduce cooling energy requirements.

However, the embodied energy of training infrastructure manufacturing GPUs, constructing data centers, and building transmission infrastructure adds a significant upstream carbon burden. Microsoft's 2025 sustainability report revealed that Scope 3 emissions (supply chain) increased by 23% in FY2024, driven primarily by AI hardware procurement and data center construction. For every tonne of operational CO_2 from AI training, an estimated 0.3-0.5 tonnes of embodied carbon may be incurred in hardware manufacturing alone.

2.2 Energy Consumption in AI Inference

While training captures public attention due to its dramatic scale, inference the process of generating responses, predictions, or outputs from trained models has emerged as the dominant energy consumer in AI systems. Current estimates indicate that inference accounts for 80–90% of total AI computing energy, a proportion expected to remain at approximately 75% through 2030 as AI features become embedded in everyday products and services.

The per-query energy consumption varies dramatically by model size, task complexity, and serving infrastructure. Google's 2025 methodology for measuring Gemini inference impacts reported median per-prompt consumption of 0.24 Wh for text prompts, 0.03 gCO_2e emissions, and 0.26 mL of water. However, these figures represent highly optimized serving infrastructure with a fleet-wide PUE of 1.09. Less efficient deployments may consume 3–10× more energy per query.

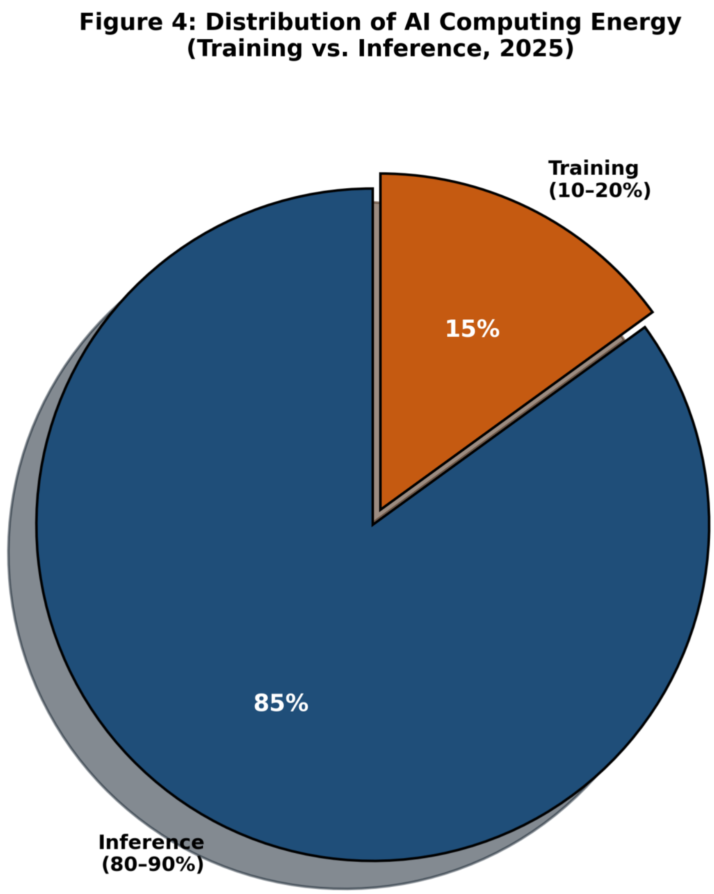

Figure 4: Distribution of AI Computing Energy Between Training and Inference Phases (2025).

Table 2: Per-Task Energy Consumption Across AI Modalities

Task Type	Model Scale	Energy per Task	CO_2 per Task
Text Generation (LLaMA 3.1 8B)	Small	~0.03 Wh	~0.01 gCO_2e
Text Generation (LLaMA 3.1 405B)	Large	~1.9 Wh	~0.6 gCO_2e
Image Generation (Stable Diffusion)	Medium	~0.6 Wh	~0.2 gCO_2e
Video Generation (5-sec clip)	Large	~0.9 kWh	~0.3 $kgCO_2e$
Voice Synthesis (1 min)	Medium	~0.15 Wh	~0.05 gCO_2e

Note: Values assume US average grid carbon intensity. Actual figures vary significantly by data center location and efficiency. Source: Google Cloud (2025); AIMultiple (2026).

The aggregate impact of inference is staggering. With billions of AI queries processed daily across ChatGPT, Google Search, Microsoft Copilot, and embedded AI features in smartphones and applications, the cumulative energy demand dwarfs training. A ChatGPT query consumes approximately 10× more electricity than a traditional Google search, and as generative AI replaces conventional search interfaces, the baseline energy cost of information retrieval rises substantially.

The IEA projects that AI-specific servers consumed 53–76 TWh in 2024 and will rise to 165–326 TWh by 2028. At the upper end of this range, AI-related electricity use alone could power approximately 22% of US households. This growth trajectory is driven not merely by query volume but by the increasing computational intensity per query as users shift from simple text generation to multimodal tasks involving images, video, and complex reasoning.

2.3 Water Consumption and Thermal Management

The water footprint of AI infrastructure has emerged as a critical but underappreciated environmental concern. Data centers generate enormous quantities of waste heat approximately 98% of electrical energy consumed is dissipated as thermal energy requiring active cooling. A 100 MW AI data center operating at full capacity can consume 1.5-3.0 million cubic meters of water annually for evaporative cooling, equivalent to the annual water consumption of 10,000–20,000 households.

The water intensity of AI workloads is substantially higher than conventional data center operations due to the concentrated heat output of GPU clusters. Google's 2025 environmental report revealed that water consumption for data center cooling increased by 17% year-over-year, reaching 9.1 billion liters in 2024. Microsoft reported a 22% increase in water consumption for its cloud and AI operations, while Meta disclosed a 15% rise. These increases substantially outpace revenue growth, indicating deteriorating water efficiency metrics.

The United Nations Regional Information Centre estimates that global AI-related water demand will reach 4.2-6.6 billion cubic meters by 2027 comparable to the annual water consumption of Denmark (5.5 billion m^3). This projection assumes continued reliance on evaporative cooling technologies; alternative approaches such as liquid immersion cooling and dry cooling could reduce water consumption by 80-95% but require significant capital investment.

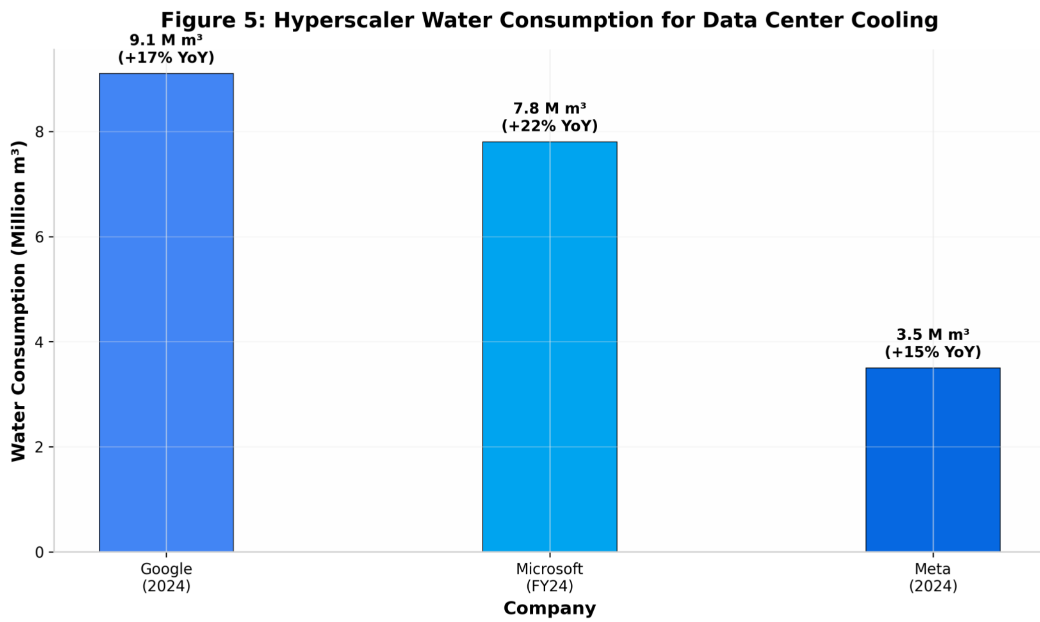

Figure 5: Hyperscaler Water Consumption for Data Center Cooling (2024). Data from corporate sustainability reports.

Table 3: Water Consumption by Major Hyperscalers and AI-Specific Projections

Company / Metric	Water Consumption (Million m³)	YoY Change (%)	Primary Cooling Method
Google (2024)	9,100	+17	Evaporative + Liquid
Microsoft (FY24)	7,800	+22	Evaporative
Meta (2024)	3,500	+15	Evaporative
Amazon AWS (Est.)	6,200	+18	Evaporative
AI-Specific (Global, 2027 Proj.)	4,200–6,600	—	Mixed

Note: AI-specific projection from UN Regional Information Centre (2025). Corporate data from annual sustainability reports.

Emerging cooling technologies offer pathways to reduce water dependence. Liquid immersion cooling submerges servers in dielectric fluid, eliminating evaporative water loss while improving heat transfer efficiency by 40–60%. Microsoft has deployed two-phase immersion cooling at its Quincy, Washington facility, achieving PUE of 1.07 with minimal water consumption. However, the dielectric fluids used (typically engineered fluorocarbons) raise concerns about persistence in the environment and potential toxicity.

Direct liquid cooling (DLC), which circulates coolant through cold plates attached to high-heat components, reduces water consumption by 60–80% compared to traditional chilled water systems while enabling higher rack densities. NVIDIA's DGX systems incorporate DLC as standard, and hyperscalers are increasingly adopting this approach for AI training clusters. The transition from air-cooled to liquid-cooled infrastructure represents a multi-billion dollar capital investment but is increasingly viewed as essential for managing the thermal loads of next-generation AI accelerators consuming 700W–1,200W per chip.

2.4 Carbon Emissions and Greenhouse Gas Impact

The carbon footprint of AI extends across three scopes of greenhouse gas emissions. Scope 1 emissions (direct combustion) are minimal for most data center operations, though backup diesel generators contribute during grid outages. Scope 2 emissions (purchased electricity) dominate the operational carbon footprint and vary by grid carbon intensity. Scope 3 emissions (supply chain) encompass hardware manufacturing, construction, and end-of-life disposal, and are increasingly recognized as representing 30–50% of total lifecycle emissions.

The carbon intensity of AI computing is fundamentally determined by the electricity grid serving the data center. Figure 3 illustrates the dramatic variation in grid carbon intensity across major data center regions. Training a model in Norway (15 gCO_2/kWh) versus South Africa (900 gCO_2/kWh) produces a 60-fold difference in emissions for identical computational workloads. This locational sensitivity has driven hyperscalers to pursue power purchase agreements (PPAs) for renewable energy, with Google, Microsoft, and Amazon collectively contracting over 50 GW of renewable capacity globally.

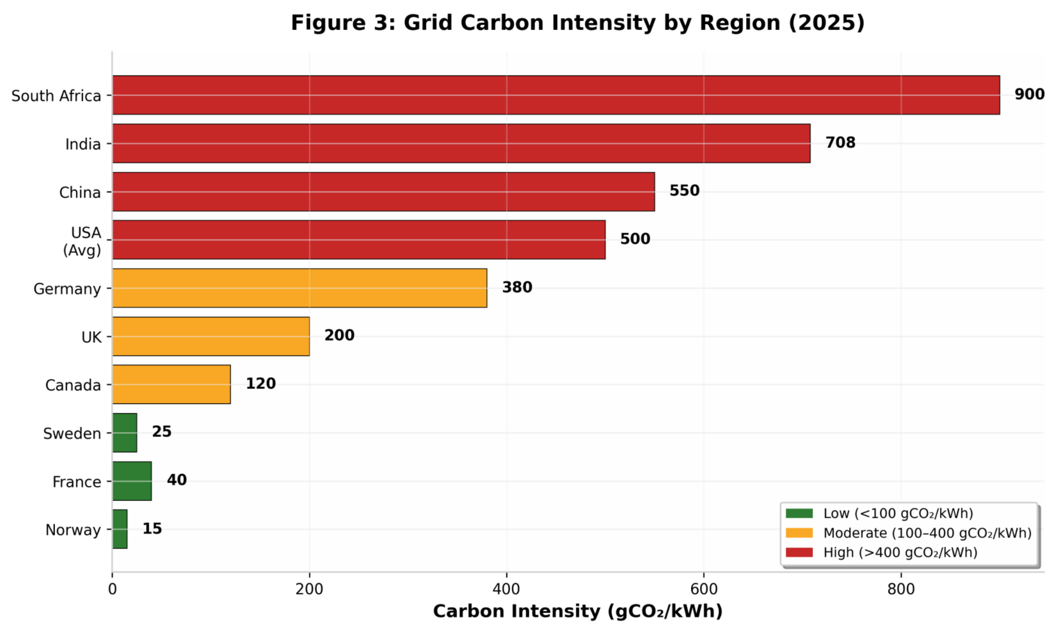

Figure 3: Grid Carbon Intensity by Region (2025). Data from Ember Global Electricity Review and IEA.

Table 4: Carbon Emissions from AI Model Training Under Different Grid Scenarios

Model	Norway Grid (15 gCO_2/kWh)	France Grid (40 gCO_2/kWh)	US Average (400 gCO_2/kWh)

GPT-3 (1,287 MWh)	19 tCO_2e	51 tCO_2e	515 tCO_2e
GPT-4 (50,000 MWh)	750 tCO_2e	2,000 tCO_2e	20,000 tCO_2e
LLaMA 3.1 405B (30,000 MWh)	450 tCO_2e	1,200 tCO_2e	12,000 tCO_2e
Annual Inference (Global, 2025)	~75,000 tCO_2e	~200,000 tCO_2e	~2,000,000 tCO_2e
Annual Inference (Global, 2030 Proj.)	~300,000 tCO_2e	~800,000 tCO_2e	~8,000,000 tCO_2e

Note: Inference estimates assume 75 TWh annual AI computing in 2025 and 300 TWh in 2030. Actual emissions depend on workload distribution across regions and real-time grid mix.

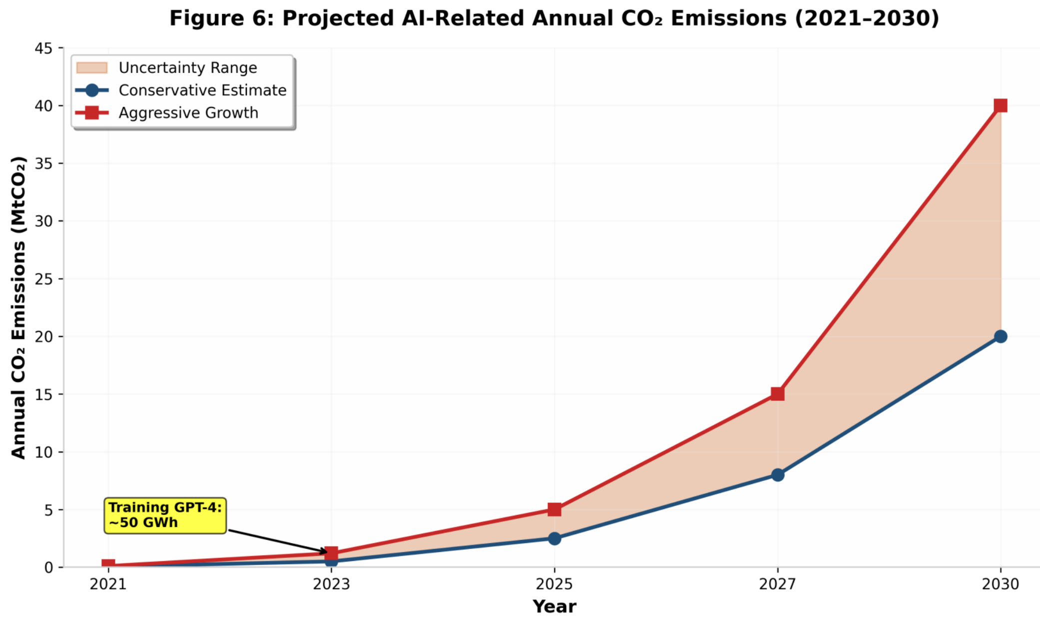

Figure 6: Projected AI-Related Annual CO_2 Emissions (2021–2030). Conservative and aggressive growth scenarios shown.

The total carbon footprint of AI in 2025 is estimated at 2–5 $MtCO_2e$ annually from direct electricity consumption, with an additional 1–2 $MtCO_2e$ from embodied emissions in hardware manufacturing. By 2030, under the IEA's base case scenario, AI-related emissions could reach 8–20 $MtCO_2e$ annually comparable to the aviation emissions of a mid-sized country such as Belgium or Sweden.

The concept of "carbon-aware computing" has emerged as a promising approach to reducing AI's emissions without reducing capability. By scheduling training jobs during periods of high renewable generation, shifting inference workloads to low-carbon regions, and dynamically scaling computational resources based on real-time grid carbon intensity, AI operators can reduce emissions by 20–40% with minimal impact on service quality. Google has implemented carbon-aware load shifting across its data center fleet, while Microsoft has committed to matching 100% of its electricity consumption with zero-carbon energy on an hourly basis by 2030.

2.5 Electronic Waste and Hardware Lifecycle

The rapid obsolescence cycle of AI hardware generates substantial electronic waste (e-waste) with significant environmental and human health implications. AI accelerators particularly GPUs have a typical useful lifespan of 3–5 years in production data centers before being retired due to performance degradation, technological obsolescence, or warranty expiration. The global data center hardware replacement cycle generates an estimated 2–3 million tonnes of e-waste annually, with AI-specific hardware representing a growing share.

NVIDIA's H100 GPUs, the dominant AI training accelerator as of 2024, contain approximately 3.5 kg of materials per unit including silicon, copper, gold, palladium, and rare earth elements. With an estimated 3.5 million H100 units shipped in 2024, the total material throughput for a single generation of AI accelerators exceeds 12,000 tonnes. The next-generation B200 and GB200 platforms increase material intensity per compute unit by 40-60%.

The environmental impact of e-waste extends beyond the hardware itself. Improper disposal of AI accelerators in developing countries where 70-80% of global e-waste is informally processed releases toxic substances including lead, mercury, cadmium, and brominated flame retardants into soil and water systems. The Basel Convention restricts transboundary movement of hazardous e-waste, but enforcement remains weak.

Circular economy approaches to AI hardware are nascent but growing. NVIDIA's certified refurbished GPU program extends hardware lifecycles by 2-3 years for inference workloads. Hyperscalers increasingly redeploy retired training GPUs to inference clusters, where lower computational intensity extends useful life. However, the specialized nature of AI accelerators limits secondary market demand compared to general-purpose computing hardware.

2.6 Rare Earth Mineral Extraction

The manufacturing of AI hardware depends upon critical minerals and rare earth elements whose extraction carries substantial environmental and social costs. A single 2 kg computer requires approximately 800 kg of raw materials including iron, copper, aluminum, silicon, and trace amounts of gallium, indium, cobalt, and neodymium. The concentration of these elements in ore bodies is typically 0.1-1%, meaning that extracting the materials for one AI accelerator may disturb 100-1,000 tonnes of earth.

Cobalt, essential for lithium-ion batteries in data center UPS systems and for certain semiconductor manufacturing processes, is predominantly sourced from the Democratic

Republic of Congo (DRC), where artisanal mining operations have been linked to child labor, deforestation, and water pollution. Approximately 70% of global cobalt production originates from the DRC, creating significant supply chain concentration risk and ethical concerns.

Gallium and indium, critical for compound semiconductors used in high-frequency AI accelerators and photonic interconnects, are byproducts of aluminum and zinc mining with limited primary production. China's dominance in gallium refining (producing 98% of global supply) creates geopolitical vulnerability and raises concerns about environmental standards in extraction and processing.

Neodymium and dysprosium, used in the permanent magnets of data center cooling fans and pump motors, are extracted through open-pit mining that generates radioactive thorium and uranium tailings. The Bayan Obo mine in Inner Mongolia, which supplies approximately 45% of global rare earth production, has created a 12 km² tailings pond containing radioactive waste that threatens groundwater resources.

Table 5 summarizes the critical minerals required for AI infrastructure and their primary environmental concerns.

Table 5: Critical Minerals in AI Infrastructure and Associated Environmental Concerns

Mineral	Application in AI	Primary Source	Key Environmental Concern
Cobalt	Batteries, semiconductors	DRC (70%)	Child labor, water pollution
Gallium	Compound semiconductors	China (98%)	Toxic byproduct processing
Indium	Transparent conductors, displays	China, Korea	Acid mine drainage
Neodymium	Permanent magnets	China (60%)	Radioactive tailings
Silicon (High-Purity)	Semiconductor wafers	China, US, Norway	Energy-intensive purification
Copper	Interconnects, power delivery	Chile, Peru, China	Acid rock drainage, habitat loss

Source: USGS Mineral Commodity Summaries (2025); IEA Critical Minerals Market Review (2025).

3. AI IN CLIMATE CHANGE MITIGATION

While the preceding sections have documented AI's substantial environmental footprint, it is equally important to recognize AI's transformative potential for climate change mitigation. The International Energy Agency estimates that AI-enabled energy system optimization could reduce global CO_2 emissions by 1.5-4.0 Gt annually by 2030 equivalent to the combined annual emissions of the European Union and Japan. This section examines the principal domains where AI contributes to climate mitigation, evaluating both demonstrated achievements and future potential.

3.1 Renewable Energy Optimization

The integration of variable renewable energy sources (wind and solar) into electrical grids presents one of the most significant technical challenges of the energy transition. AI addresses this challenge through multiple mechanisms:

Forecasting Accuracy: Machine learning models improve wind power forecasting by 15–30% and solar forecasting by 20–25% compared to traditional numerical weather prediction methods. Google's DeepMind applied neural networks to wind farm output prediction, achieving 36% improvement in forecast accuracy and increasing the value of wind energy by approximately 20%. These improvements enable grid operators to reduce spinning reserves (backup fossil fuel generation) and increase renewable penetration.

Predictive Maintenance: AI-powered condition monitoring of wind turbines and solar inverters reduces unplanned downtime by 30–50%. Vibration analysis, thermal imaging, and acoustic sensors feed machine learning models that predict component failures 2–6 weeks in advance, enabling proactive maintenance scheduling that maximizes capacity factors and extends asset lifespans.

Resource Assessment: Deep learning models trained on satellite imagery, meteorological data, and historical generation records identify optimal sites for renewable energy development with 40% greater accuracy than conventional GIS-based methods. This reduces development risk, accelerates project timelines, and minimizes land-use conflicts by avoiding environmentally sensitive areas.

Grid Integration: Reinforcement learning algorithms optimize the dispatch of hybrid renewable-storage systems, balancing generation, storage, and demand in real-time. Tesla's Autobidder platform uses AI to manage over 5.5 GWh of battery storage across multiple markets, generating revenue from frequency regulation and arbitrage while supporting grid stability.

The global impact of AI on renewable energy is substantial but difficult to quantify precisely. The IEA estimates that AI-enabled improvements in forecasting and grid management could facilitate an additional 200–400 TWh of renewable generation by 2030, avoiding 100–200 $MtCO_2e$ of fossil fuel emissions.

3.2 Smart Grid Management

Electrical transmission and distribution systems represent a critical infrastructure domain where AI delivers immediate climate benefits. Global transmission and distribution losses average 8–15% of generated electricity, representing approximately 1,500 TWh of wasted energy and 750 $MtCO_2e$ of avoidable emissions annually.

AI-powered smart grid applications reduce these losses through several mechanisms:

Load Forecasting: Neural network models predict electricity demand at 15-minute to hourly intervals with 95%+ accuracy, enabling utilities to optimize generation dispatch and reduce overproduction. The California Independent System Operator (CAISO) has implemented AI-driven demand forecasting that has reduced reserve margins by 3–5%, deferring construction of peaker plants.

Anomaly Detection: Machine learning algorithms identify transformer failures, line faults, and unauthorized consumption with 90%+ accuracy, reducing outage duration and associated emissions from backup generation. Duke Energy's AI-powered grid monitoring system has reduced outage restoration times by 25% across its service territory.

Voltage Optimization: Reinforcement learning controllers adjust distribution network voltages in real-time based on load conditions, reducing resistive losses by 2–4%. National Grid's voltage optimization program, powered by AI, saves approximately 100 GWh annually across the UK network.

Demand Response: AI-enabled demand response programs aggregate flexible loads (industrial processes, EV charging, building HVAC systems) to provide grid services traditionally supplied by fossil fuel peaker plants. OhmConnect's AI platform has dispatched over 1 GWh of demand response in California, avoiding the activation of gas-fired peaker plants during grid stress events.

The economic case for AI in grid management is compelling. The US Department of Energy estimates that full deployment of AI-enabled grid modernization could save \$100–200 billion in infrastructure investments over the next decade while reducing emissions by 200–400 $MtCO_2e$ annually.

3.3 Carbon Capture and Storage

Carbon capture, utilization, and storage (CCUS) technologies are essential for decarbonizing heavy industry and achieving net-zero emissions, but they face significant technical and economic barriers. AI accelerates CCUS development through materials discovery, process optimization, and site selection.

Materials Discovery: Deep learning models predict the CO_2 adsorption capacity and selectivity of novel metal-organic frameworks (MOFs) and zeolites with 85%+ accuracy, reducing the experimental screening burden by 90%. Researchers at MIT and Berkeley have used AI to identify over 50 promising new sorbent materials in silico, several of which have been validated experimentally with performance exceeding commercial benchmarks.

Process Optimization: Reinforcement learning controllers optimize solvent regeneration cycles in post-combustion capture plants, reducing energy consumption by 15–25%. The Petra Nova facility in Texas, enhanced with AI process control, achieved 90% capture efficiency with 30% lower energy penalty than design specifications.

Site Selection and Monitoring: Machine learning models trained on seismic, geologic, and hydrologic data identify optimal CO_2 storage formations with reduced risk of leakage. AI-powered seismic monitoring detects microseismic events associated with CO_2 plume migration, providing early warning of potential containment breaches.

Direct Air Capture (DAC): AI optimizes the design and operation of DAC systems, which must process enormous volumes of air to extract dilute CO_2 (400 ppm). Climeworks' Orca facility in Iceland uses AI to optimize fan speeds, temperature profiles, and sorbent regeneration cycles, achieving energy consumption of 2,500–3,000 kWh per tonne of CO_2 captured.

While CCUS remains expensive ($50–200 per tonne CO_2), AI-driven improvements in materials and process efficiency could reduce costs to $30–50 per tonne by 2035, making CCUS economically viable for hard-to-abate sectors including cement, steel, and chemicals production.

3.4 Precision Agriculture

Agriculture contributes approximately 10–12% of global greenhouse gas emissions, primarily through methane from livestock, nitrous oxide from fertilizer application, and CO_2 from land-use change. AI-enabled precision agriculture offers pathways to reduce these emissions while maintaining or improving yields.

Variable Rate Application: Machine learning models analyze soil sensors, satellite imagery, and weather data to optimize fertilizer and irrigation application at sub-field resolution. John Deere's AI-powered See & Spray technology reduces herbicide use by 70–90% by targeting individual weeds rather than broadcasting chemicals across entire fields. Bayer's Climate FieldView platform uses AI to optimize nitrogen application, reducing fertilizer overuse by 15–20% and associated N_2O emissions by 10–15%.

Yield Prediction and Crop Selection: Deep learning models predict crop yields with 90%+ accuracy 2–4 weeks before harvest, enabling farmers to optimize harvest timing, storage, and market decisions. AI-driven crop recommendation systems suggest optimal crop rotations and varieties based on soil conditions, climate projections, and market prices, improving land-use efficiency.

Livestock Management: Computer vision and sensor-based AI systems monitor animal health, behavior, and feed efficiency, reducing methane emissions per unit of production by 10–15%. Precision livestock feeding algorithms optimize ration composition to minimize enteric fermentation while maintaining nutritional requirements.

Supply Chain Optimization: AI-powered logistics platforms reduce food waste a major source of emissions when decomposing food releases methane in landfills by 20–30% through improved demand forecasting, inventory management, and routing optimization. Walmart's AI-driven fresh food management system has reduced food waste by 50% in pilot stores.

The aggregate climate impact of precision agriculture is substantial. The World Economic Forum estimates that AI-enabled agricultural optimization could reduce global agricultural emissions by 300–500 $MtCO_2e$ annually by 2030 while improving food security through yield increases of 10–15%.

3.5 Climate Modeling and Prediction

AI is revolutionizing climate science by enabling higher-resolution models, faster simulations, and improved prediction of extreme events. Traditional climate models require months of supercomputing time to simulate decades of climate evolution, limiting the number of scenarios that can be explored and the spatial resolution of outputs.

Neural Network Emulators: Deep learning models trained on traditional climate simulation outputs can generate equivalent predictions 1,000–10,000× faster, enabling ensemble forecasting with thousands of scenarios rather than dozens. Google's GraphCast model predicts weather variables up to 10 days in advance with accuracy exceeding traditional numerical weather prediction at 99.7% lower computational cost.

Downscaling: AI models downscale coarse-resolution climate projections (100–300 km grid cells) to local scales (1–10 km) needed for impact assessment and adaptation planning. These downscaled projections inform infrastructure design, agricultural planning, and public health preparedness with unprecedented spatial detail.

Extreme Event Prediction: Machine learning models identify precursors to extreme weather events heatwaves, droughts, floods, tropical cyclones with lead times 2-5 days longer than traditional methods. Google's flood forecasting initiative provides 7-day advance warnings for 80+ countries, directly enabling evacuation and preparedness measures that save lives and reduce disaster-related emissions from emergency response.

Carbon Cycle Modeling: AI improves representation of carbon cycle processes in Earth system models, including photosynthesis, respiration, and ocean carbon uptake. These improvements reduce uncertainty in climate sensitivity estimates and carbon budget calculations, informing more precise emissions targets and mitigation strategies.

3.6 Transportation and Logistics Optimization

Transportation accounts for approximately 16% of global greenhouse gas emissions, with road transport representing three-quarters of this total. AI optimizes transportation systems through route optimization, traffic management, autonomous driving, and modal shift facilitation.

Route Optimization: Machine learning algorithms solve vehicle routing problems with 15–30% improvement over heuristic methods, reducing fuel consumption and emissions for logistics fleets. UPS's ORION system, powered by AI, has saved the company 10 million gallons of fuel annually by optimizing delivery routes to minimize left turns and idling time.

Traffic Management: AI-powered adaptive traffic signal control reduces urban congestion by 15–25%, cutting vehicle emissions by 10–20% in managed corridors. Pittsburgh's Surtrac system uses AI to optimize traffic signals in real-time, reducing travel times by 25% and emissions by 21% across the pilot network.

Autonomous Vehicles: AI-powered advanced driver assistance systems (ADAS) improve fuel efficiency by 5–10% through optimized acceleration, braking, and speed maintenance. Tesla's Autopilot system reduces energy consumption per mile by approximately 8% through predictive speed optimization and regenerative braking management.

Modal Shift: AI-powered multimodal journey planners facilitate shifts from private vehicles to public transit, cycling, and walking by providing integrated, real-time information on options, durations, and emissions. Google Maps' eco-friendly routing features have been shown to increase sustainable mode share by 5–10% among users.

Aviation and Shipping: AI optimizes flight paths to minimize fuel consumption and contrail formation (which accounts for approximately 35% of aviation's climate impact). Google Flights' AI-powered contrail avoidance system has reduced contrail-forming flight paths by 50% in pilot programs. In maritime shipping, AI route optimization reduces fuel consumption by 5–15%.

The IEA estimates that AI-enabled transportation optimization could reduce global transport emissions by 500–1,000 $MtCO_2e$ annually by 2030, with the greatest potential in urban logistics and freight transportation.

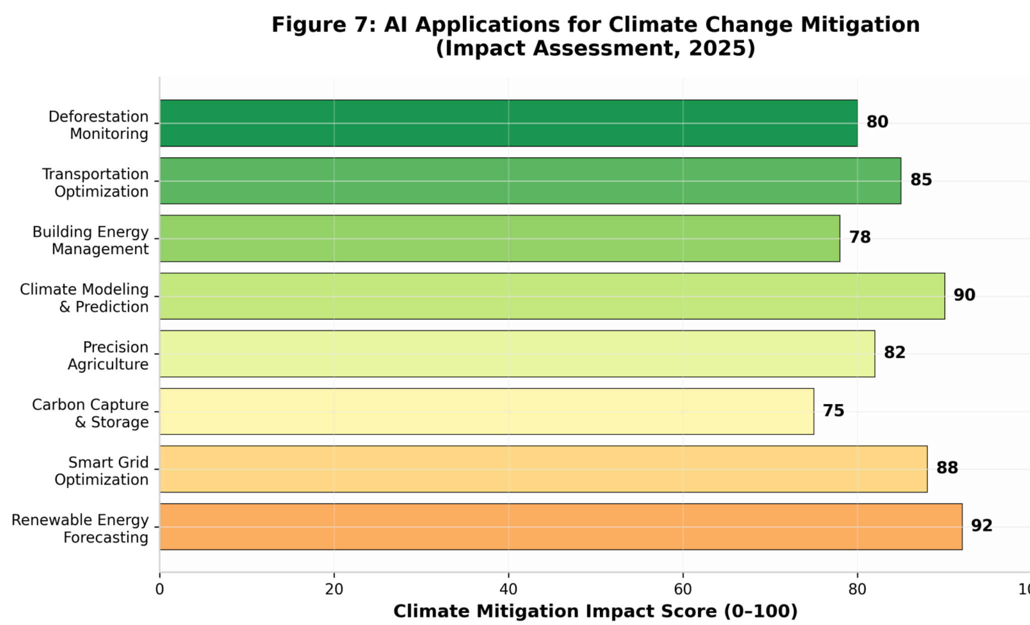

Figure 7: AI Applications for Climate Change Mitigation—Impact Assessment (2025). Scores based on emission reduction potential, scalability, and technology readiness.

4. THE JEVONS PARADOX AND REBOUND EFFECTS

4.1 Efficiency Gains vs. Demand Growth

The Jevons Paradox, first observed by William Stanley Jevons in 1865 regarding coal consumption, posits that improvements in resource efficiency can increase total resource consumption by lowering costs and stimulating new applications. In the context of AI, this paradox manifests as computational efficiency gains (faster chips, optimized algorithms, better utilization) being offset or more than offset by increased deployment, larger models, and novel applications that would not have been economically viable at higher computational costs.

The evidence for a Jevons effect in AI is compelling. Between 2012 and 2024, the energy efficiency of AI training (measured as energy per floating-point operation) improved by approximately 10,000× due to advances in GPU architecture, model parallelism, and algorithmic optimization. Yet during the same period, total AI training energy consumption increased by approximately 300,000×, as efficiency gains enabled the training of vastly larger models on exponentially more data. The net effect has been a 30× increase in total training energy despite dramatic per-computation efficiency improvements.

Figure 8 illustrates this dynamic for AI inference. While energy per query has declined by 50–70% annually due to model distillation, quantization, and hardware improvements, total AI query volume has grown by 100-300% annually, driven by the integration of AI into search engines, productivity software, smartphones, and consumer applications. The result is a net increase in total AI energy consumption despite and partially because of efficiency improvements.

Several mechanisms drive the Jevons effect in AI:

Direct Rebound: Lower computational costs enable existing applications to process more data, run more frequently, or serve more users. A company that previously ran weekly batch analytics may shift to real-time processing; a search engine that previously returned 10 results may now generate a full AI-composed response.

Indirect Rebound: AI efficiency gains reduce the cost of AI-powered products and services, expanding market size and adoption. As AI-generated content becomes cheaper than human-created content, demand for content creation increases dramatically potentially increasing total resource consumption even as per-unit costs fall.

Economy-Wide Rebound: AI-driven productivity improvements across the economy increase overall economic output and energy demand. While AI optimizes individual processes, the resulting economic growth stimulates energy consumption in other sectors a macro-level rebound effect that may partially or fully offset the direct savings.

The magnitude of rebound effects in AI remains uncertain. Estimates from the literature suggest that rebound effects in information and communication technologies (ICT) range from 20% to 80% of direct efficiency savings, with AI-specific rebound potentially at the higher end due to its transformative nature and rapid adoption. If rebound effects exceed 100%, AI efficiency improvements result in net increases in energy consumption and emissions a "backfire" effect.

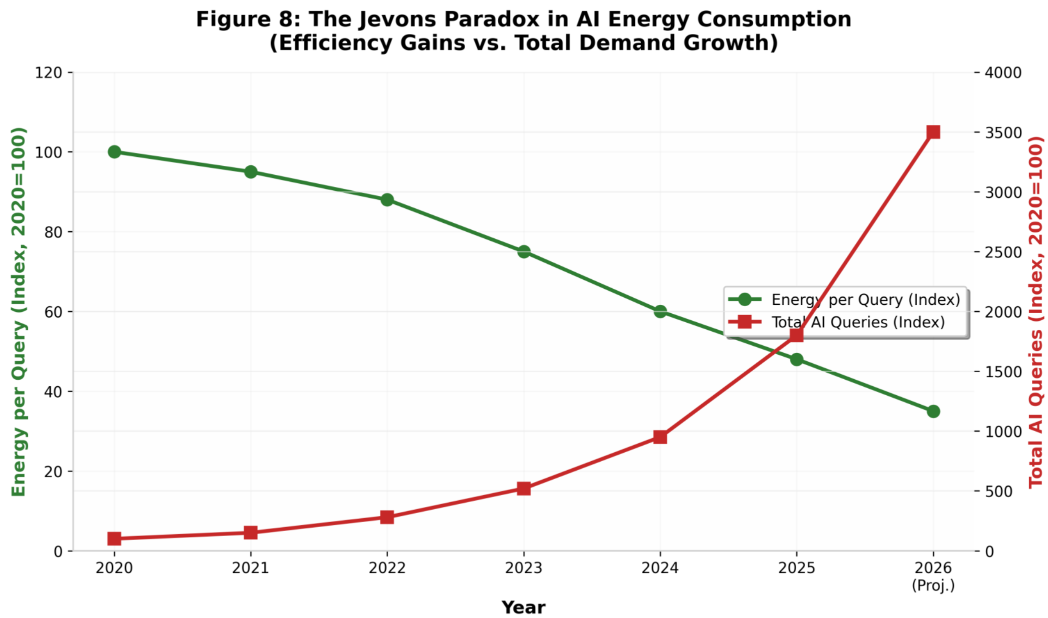

Figure 8: The Jevons Paradox in AI Energy Consumption Efficiency Gains vs. Total Demand Growth (2020–2026).

4.2 Case Studies in AI-Induced Demand

Several case studies illustrate the Jevons effect in practice:

Case Study 1: Generative AI in Search. Google's integration of Gemini into Search results increased per-query energy consumption by approximately 10× compared to traditional search. While Google has implemented efficiency optimizations, the total energy impact depends on the fraction of queries routed through generative AI. If 50% of Google's 8.5 billion daily queries shift to AI-enhanced responses, the additional energy consumption would equal approximately 15 TWh annually equivalent to the electricity consumption of 1.4 million US households.

Case Study 2: AI-Generated Content. The proliferation of AI image and video generation tools (DALL-E, Midjourney, Sora, Runway) has created a new category of computational demand. A single AI-generated image consumes 0.5-2 kWh, while a 5-second AI video clip may consume 0.5–2 kWh. With millions of images and videos generated daily, the aggregate energy consumption is substantial and growing rapidly as quality and resolution improve.

Case Study 3: Autonomous Vehicles. While autonomous driving promises fuel efficiency improvements of 10-30% through optimized routing and driving behavior, the onboard AI computing platforms consume 1-5 kW continuously during operation. For a vehicle traveling 20,000 km annually, this adds 200-1,000 kWh of computational energy consumption potentially offsetting a significant fraction of the drivetrain efficiency gains, particularly for electric vehicles where computational load directly reduces range.

Case Study 4: Smart Buildings. AI-powered building management systems reduce HVAC energy consumption by 15-30% through predictive control and occupancy optimization. However, the proliferation of AI-enabled sensors, cameras, and edge computing devices increases the building's baseline electricity consumption. For a typical commercial building, the net energy savings remain positive (10-20%), but the margin is narrower than direct efficiency metrics suggest.

These case studies demonstrate that AI's net environmental impact cannot be assessed through efficiency metrics alone. The interaction between efficiency, adoption, and application scope determines whether AI delivers net environmental benefits or exacerbates resource consumption.

5. REGULATORY FRAMEWORKS AND POLICY RESPONSES

5.1 European Union Regulations

The European Union has emerged as the global leader in AI environmental regulation, integrating sustainability requirements into its comprehensive AI governance framework. The EU AI Act, adopted in 2024 and entering into force in 2025, includes provisions for environmental impact assessment of high-risk AI systems, though the primary focus remains on safety and fundamental rights rather than climate.

More directly relevant is the EU's Green Deal and associated legislation. The Energy Efficiency Directive (recast 2023) requires data centers with power consumption exceeding 500 kW to report energy consumption, waste heat recovery potential, and water usage to national authorities. The European Sustainability Reporting Standards (ESRS) mandate disclosure of Scope 1, 2, and 3 emissions for large companies, including AI infrastructure providers.

The proposed EU Data Center Regulation (under development as of 2026) would establish minimum efficiency standards, require waste heat utilization where technically feasible, and mandate renewable energy procurement for new data centers. France has taken a particularly proactive approach, requiring data centers to report Power Usage Effectiveness (PUE) metrics and carbon intensity, and providing tax incentives for facilities achieving PUE below 1.2.

The EU's Carbon Border Adjustment Mechanism (CBAM), phased in from 2026, will indirectly affect AI hardware manufacturing by imposing carbon costs on imports of steel, aluminum, and cement used in data center construction. This creates incentives for domestic manufacturing using renewable energy and for supply chain decarbonization.

Table 6 summarizes key EU regulatory instruments affecting AI environmental impact.

Table 6: Key EU Regulatory Instruments for AI Environmental Governance

Regulation	Scope	Key Requirements	Status
EU AI Act	High-risk AI systems	Risk assessment, transparency	In force (2025)
Energy Efficiency Directive	Data centers >500 kW	Energy/water reporting	In force (2023)
ESRS Standards	Large companies	Scope 1/2/3 disclosure	In force (2024)
Data Center Regulation (Prop.)	All data centers	PUE standards, heat recovery	Under development
CBAM	Import of carbon-intensive goods	Carbon border tariffs	Phased in (2026)

5.2 United States Policy Landscape

The United States has adopted a less prescriptive, more market-oriented approach to AI environmental regulation, though recent developments suggest increasing federal attention to data center sustainability.

At the federal level, the Department of Energy's Better Buildings Initiative includes data center efficiency programs that provide technical assistance and recognition for facilities achieving PUE below 1.4. The Energy Act of 2020 directed DOE to study data center energy efficiency, resulting in the 2023 report that documented the 4.4% share of national electricity consumption and projected continued rapid growth.

The Biden Administration's Executive Order on Safe, Secure, and Trustworthy AI (October 2023) directed federal agencies to assess AI's environmental impacts and develop strategies for sustainable AI development. The National Institute of Standards and Technology (NIST) is developing an AI Risk Management Framework that includes environmental sustainability as a consideration, though the framework remains voluntary.

Several states have implemented more aggressive policies. California's SB 100 requires 100% zero-carbon electricity by 2045, indirectly driving data center decarbonization. Virginia, which hosts approximately 35% of US data center capacity, has faced growing pressure to address the strain on electrical infrastructure, with Dominion Energy projecting that data centers could consume 40% of Virginia's electricity by 2035.

The Inflation Reduction Act (2022) provides substantial tax credits for renewable energy and energy storage projects that indirectly benefit data center decarbonization. The 45Q tax credit for carbon capture and the 48C clean energy manufacturing credit support technologies relevant to sustainable AI infrastructure.

Industry self-regulation has partially filled the policy gap. The Sustainable IT Coalition, comprising Microsoft, Google, Amazon, and Meta, has committed to net-zero emissions by 2030 and to publishing standardized environmental impact metrics for AI services. However, the voluntary nature of these commitments limits their effectiveness.

5.3 China and Asia-Pacific Initiatives

China, as the world's largest consumer of electricity and the second-largest data center market, has implemented increasingly stringent policies on data center energy efficiency and carbon emissions. The National Development and Reform Commission's (NDRC) "Guidelines on the Development of Data Centers" (2021) established PUE targets of 1.3 for new data centers in eastern regions and 1.25 for western regions, where cooler climates and abundant renewable energy reduce cooling and grid carbon intensity.

The "East Data West Computing" (东数西算) initiative, launched in 2022, strategically directs AI training workloads to western provinces (Inner Mongolia, Guizhou, Gansu, Ningxia) where renewable energy is abundant and land costs are lower, while reserving eastern data centers for latency-sensitive inference workloads. This policy explicitly recognizes the locational determinants of AI's carbon footprint and represents the most geographically explicit AI sustainability policy globally.

China's carbon neutrality commitment (2060 target) and the national emissions trading scheme (ETS), which covers power generation, create economic incentives for data center operators to reduce emissions. However, enforcement of ETS compliance for data centers remains limited, and the carbon price (approximately $10-15 per tonne CO_2e as of 2025) is too low to significantly influence operational decisions.

Singapore, despite its limited land area and tropical climate, has emerged as a major data center hub. The government imposed a moratorium on new data center construction in 2019–2022 due to energy constraints, then lifted the moratorium with strict requirements for green building certification, PUE below 1.3, and renewable energy procurement. Japan's Green Transformation (GX) initiative includes data center efficiency standards and subsidies for renewable energy integration. South Korea's Green New Deal provides funding for data center cooling innovation and renewable energy deployment.

5.4 International Standards and Guidelines

International standards bodies have begun developing frameworks for measuring and reporting AI environmental impacts, though standardization remains in early stages.

The International Organization for Standardization (ISO) has initiated work on ISO/IEC 22989, which includes provisions for sustainability assessment of AI systems. The Green Software Foundation, a Linux Foundation project supported by Microsoft, Google, and Accenture, has developed the Software Carbon Intensity (SCI) specification a methodology for calculating the carbon emissions of software applications including AI services.

The IEEE Standards Association is developing P2857, a standard for measuring the energy efficiency of machine learning workloads. This standard aims to provide consistent benchmarking methodologies that enable comparison across models, hardware platforms, and deployment scenarios a critical prerequisite for regulatory standard-setting and market-based incentives.

The UN Framework Convention on Climate Change (UNFCCC) has not yet developed specific guidance on AI emissions accounting, but the Global Digital Compact (adopted at the 2024 Summit of the Future) includes commitments to "ensure that digital technologies, including AI, contribute to climate action and do not exacerbate environmental degradation." The compact calls for standardized environmental impact reporting for digital infrastructure and for integrating AI emissions into national climate commitments (NDCs).

The OECD AI Principles (2019, updated 2024) include sustainability as a cross-cutting consideration, recommending that AI systems be designed and deployed to minimize environmental impact. However, the principles remain non-binding, and implementation varies widely across member states.

The absence of harmonized international standards for AI environmental accounting creates significant challenges for cross-border comparison, regulatory enforcement, and corporate reporting. The development of international standards ideally under the auspices of ISO or a new specialized body represents a critical priority for effective global governance of AI's environmental impact.

6. SUSTAINABLE AI: STRATEGIES AND BEST PRACTICES

6.1 Green Data Center Design

The design and operation of data centers fundamentally determines the environmental footprint of AI computing. Green data center design integrates energy efficiency, renewable energy, water conservation, and circular economy principles to minimize environmental impact while maintaining computational performance.

Power Usage Effectiveness (PUE) the ratio of total facility energy to IT equipment energy remains the primary metric for data center efficiency. The industry average PUE has declined from approximately 2.0 in 2007 to 1.5–1.6 in 2024, with hyperscale facilities achieving 1.1–1.2 through advanced cooling, waste heat recovery, and optimized power distribution. Google's fleet-wide PUE of 1.10 and Microsoft's average of 1.12 represent current best practice, though these figures exclude embodied energy and water consumption.

Free cooling utilizing ambient air or water for cooling without mechanical refrigeration reduces cooling energy by 30-60% in temperate climates. Google's data center in Hamina, Finland, uses seawater from the Baltic Sea for cooling, achieving PUE below 1.09 with minimal water consumption. Facebook's (Meta) data center in Luleå, Sweden, leverages Arctic air for 100% free cooling, operating at PUE of 1.04 during winter months.

Waste heat recovery transforms data centers from energy consumers to energy producers. In Denmark, Meta's Odense data center recovers waste heat to warm 11,000 homes through the district heating network. In Finland, Yandex's Mäntsälä data center provides heat to 20,000 households. These examples demonstrate that data centers can become net-positive environmental assets when integrated into urban energy systems.

Renewable energy procurement has become standard practice for hyperscalers, with Google, Microsoft, Amazon, and Meta collectively contracting over 50 GW of renewable capacity. However, the quality of renewable energy procurement varies significantly. Annual matching (matching total annual consumption with renewable generation) is the most common approach but does not ensure that AI workloads run on clean energy in real-time. 24/7 carbon-free energy matching represents the gold standard but is technically and economically challenging.

Table 7 compares data center efficiency metrics across major operators.

Table 7: Data Center Efficiency Metrics by Major Operator (2024)

Operator	Fleet PUE	Renewable Energy (% of consumption)	Water Efficiency (L/kWh)	Waste Heat Recovery
Google	1.10	100% (annual match)	0.15	Limited pilots
Microsoft	1.12	100% (annual match)	0.18	Denmark, Finland
Amazon AWS	1.13	90%	0.20	Limited
Meta	1.09	100% (annual match)	0.14	Denmark, Ireland
Industry Average	1.55	35%	0.45	<5%

Note: PUE and water efficiency data from corporate sustainability reports. Industry average from Uptime Institute Global Data Center Survey (2024).

6.2 Carbon-Aware Computing

Carbon-aware computing represents a paradigm shift from optimizing for computational efficiency alone to optimizing for carbon efficiency the minimization of CO_2 emissions per unit of useful computation. This approach recognizes that the carbon intensity of electricity varies by hour, day, and season, and that scheduling computational workloads to coincide with low-carbon generation can reduce emissions without reducing computational output.

Google pioneered carbon-aware computing at scale, implementing a system that shifts flexible workloads (training jobs, data processing, video rendering) to times and locations with low grid carbon intensity. In 2023, Google shifted over 5 TWh of compute workload to low-carbon periods, avoiding an estimated 1.5 $MtCO_2e$ of emissions. The system uses real-time grid carbon intensity data from electricityMap and proprietary forecasting models to predict optimal scheduling windows 24–48 hours in advance.

Microsoft's carbon-aware computing initiative extends this approach to Azure cloud services, offering customers the option to schedule workloads during periods of low grid carbon intensity. The "sustainability calculator" provides real-time visibility into the carbon footprint of cloud resources, enabling informed decisions about workload timing, location, and resource allocation.

The technical implementation of carbon-aware computing requires:

Temporal Flexibility: Workloads must be delay-tolerant to shift across hours or days. Training jobs, batch processing, and model fine-tuning are highly suitable; real-time inference and latency-sensitive applications are not.

Spatial Flexibility: Data and models must be replicable across geographic regions with different grid carbon intensities. This requires robust data synchronization, model versioning, and regulatory compliance across jurisdictions.

Predictive Capability: Accurate forecasting of grid carbon intensity 12–48 hours ahead enables proactive scheduling. Machine learning models trained on historical generation mix, weather forecasts, and demand patterns achieve 85–95% accuracy in carbon intensity prediction.

The potential emission reductions from carbon-aware computing are substantial. Analysis by the Electric Power Research Institute (EPRI) suggests that widespread adoption of carbon-aware scheduling in US data centers could reduce AI-related emissions by 15–25% with minimal impact on service quality. For delay-tolerant workloads, reductions of 40–60% are achievable by shifting from high-carbon evening peaks to low-carbon midday solar generation or overnight wind generation.

6.3 Model Efficiency and Optimization

The computational cost of AI models has grown exponentially with model size, but algorithmic innovations offer pathways to maintain capability while reducing resource requirements. Model efficiency techniques can reduce energy consumption by 50–99% with minimal impact on task performance.

Model Distillation transfers knowledge from large, complex "teacher" models to smaller "student" models. Google's DistilBERT achieves 97% of BERT's performance on natural language understanding tasks with 60% fewer parameters and 40% lower inference energy. More recent distillation techniques have achieved 95% performance retention with 90% parameter reduction for large language models.

Quantization reduces the numerical precision of model weights from 32-bit floating point to 8-bit or 4-bit integers, reducing memory bandwidth requirements and enabling faster computation on specialized hardware. INT8 quantization typically reduces model size by 4× and inference energy by 2–4× with <1% accuracy loss. NVIDIA's TensorRT and Qualcomm's AI Stack support INT4 quantization for edge deployment, achieving 8× size reduction.

Pruning removes redundant or low-importance connections from neural networks, reducing parameter counts by 50–90% with minimal performance degradation. Structured pruning (removing entire neurons or channels) enables hardware-friendly sparse computation, while unstructured pruning requires specialized sparse matrix operations.

Neural Architecture Search (NAS) uses AI to discover efficient model architectures optimized for specific hardware platforms and energy constraints. Google's EfficientNet, discovered through NAS, achieves state-of-the-art image classification accuracy with 8.4× fewer parameters and 16× lower FLOPs than previous models. More recent NAS techniques incorporate carbon cost as an optimization objective, directly minimizing training emissions.

Mixture of Experts (MoE) architectures activate only a subset of model parameters for each input, reducing per-token computation while maintaining large model capacity. Google's Gemini 1.5 and Mixtral 8x7B use MoE to achieve performance comparable to dense models with 3–5× lower inference energy.

The cumulative impact of these techniques is substantial. A recent study by the Allen Institute for AI demonstrated that combining distillation, quantization, and pruning reduced the energy cost of a question-answering task by 99.7% while maintaining 95% of the original model's accuracy. However, these efficiency gains must be weighed against the energy cost of developing and validating optimized models a consideration often overlooked in efficiency analyses.

6.4 Hardware Innovation

Hardware innovation represents the most direct pathway to reducing AI's energy consumption. The transition from general-purpose CPUs to specialized AI accelerators has already improved energy efficiency by 10–100× for deep learning workloads, and continued innovation promises further gains.

GPUs (Graphics Processing Units) remain the dominant AI training platform, with NVIDIA's H100 and emerging B200 platforms representing the state of the art. The H100 achieves approximately 3× the training throughput of its predecessor (A100) at comparable power consumption, effectively halving energy per training run. The B200, announced in 2024, further improves efficiency through multi-chip modules and enhanced Tensor Core architectures.

TPUs (Tensor Processing Units), developed by Google for internal use and cloud deployment, offer competitive efficiency for specific workloads. TPU v5p achieves approximately 459 TFLOPS per chip at 600W, with specialized support for transformer architectures that dominate current AI models.

ASICs (Application-Specific Integrated Circuits) represent the most energy-efficient approach for fixed workloads. Cerebras' wafer-scale engine (WSE-3) integrates 4 trillion transistors on a single wafer, achieving unprecedented parallelism for sparse AI workloads. SambaNova's Reconfigurable Dataflow Unit (RDU) adapts hardware configuration to specific models, improving efficiency by 2–5× compared to GPUs for certain workloads.

Neuromorphic computing, inspired by biological neural networks, offers radical efficiency improvements for specific AI tasks. Intel's Loihi 2 chip and IBM's NorthPole processor demonstrate 100–1,000× energy efficiency improvements for inference on spiking neural networks, though these architectures remain limited to specialized applications.

Photonic computing, which uses light rather than electrons for computation, promises 10–100× efficiency improvements by eliminating resistive losses in interconnects and computation. Lightmatter's Passage photonic interconnect and Ayar Labs' optical I/O solutions are approaching commercial deployment, with potential to reduce the energy cost of data movement a dominant contributor to AI energy consumption by 90%.

The hardware innovation pipeline is robust, with multiple promising technologies in development. However, the 3-5 year lag between research and commercial deployment means that near-term AI energy growth will be constrained by currently available hardware. The IEA projects that hardware efficiency improvements will reduce AI energy per computation by 30-50% by 2030, but this will be partially offset by increased computational demand from larger models and broader deployment.

6.5 Circular Economy for AI Hardware

The circular economy approach to AI hardware aims to minimize resource extraction, maximize product lifespans, and ensure responsible end-of-life management. This approach addresses the embodied carbon and e-waste challenges documented in Sections 2.5 and 2.6.

Design for Longevity: Extending hardware useful life from 3–5 years to 7–10 years reduces embodied carbon per unit of computation by 30–50%. Modular designs that enable component-level upgrades (memory, storage, networking) rather than full system replacement support longevity. Google's custom-designed data center servers incorporate modular components that can be upgraded independently, extending chassis lifespans to 8+ years.

Refurbishment and Reuse: Retired training GPUs can be redeployed for inference workloads where lower computational intensity extends useful life. NVIDIA's certified refurbished program validates second-hand GPUs for continued operation, and hyperscalers increasingly operate internal "cascade" systems that migrate hardware from training to inference to storage tiers over its lifecycle.

Recycling and Material Recovery: AI hardware contains valuable materials (gold, palladium, copper, rare earth elements) that can be recovered through specialized recycling processes. Urban mining of e-waste recovers 40–60% of contained metals, compared to 1–5% recovery rates for primary ore extraction. However, current recycling infrastructure is inadequate for the scale of AI hardware retirement, with only 17.4% of global e-waste formally recycled as of 2024.

Extended Producer Responsibility (EPR): The EU's WEEE Directive and similar regulations in Japan and South Korea require electronics manufacturers to finance end-of-life collection and recycling. Extending EPR to AI accelerators and data center equipment would create financial incentives for design-for-recycling and establish funding for recycling infrastructure development.

Supply Chain Transparency: Blockchain and AI-powered supply chain tracking systems enable verification of responsible sourcing for critical minerals. The Responsible Minerals Initiative (RMI) has developed standards for cobalt, tantalum, tin, and tungsten sourcing that are increasingly adopted by electronics manufacturers. Expansion of these standards to gallium, indium, and other AI-critical materials would address the supply chain concerns documented in Section 2.6.

The economic viability of circular economy approaches for AI hardware depends on several factors: the residual value of retired components, the cost of refurbishment versus new purchase, regulatory requirements for responsible disposal, and consumer willingness to purchase refurbished equipment. Current market conditions favor new hardware purchases due to rapid performance improvements, but as Moore's Law slows and efficiency gains diminish, the economic case for circularity strengthens.

7. FUTURE OUTLOOK AND RECOMMENDATIONS

7.1 Projected Trends (2026–2035)

The trajectory of AI's environmental impact over the next decade will be shaped by the interplay of technological innovation, market dynamics, regulatory pressure, and societal choices. Three scenarios illustrate the range of possible outcomes:

Conservative Growth Scenario: Under this scenario, AI adoption proceeds at a measured pace, efficiency improvements keep pace with demand growth, and regulatory frameworks successfully constrain environmental externalities. AI-specific electricity consumption reaches 200–250 TWh by 2030 and stabilizes at 300–400 TWh by 2035. Carbon emissions peak at 10–15 $MtCO_2e$ annually before declining as grid decarbonization outpaces demand growth. Water consumption grows to 6–8 billion m^3 annually but is increasingly met through recycled and non-potable sources.

Base Case Scenario: This scenario assumes continued rapid AI adoption with moderate efficiency improvements and partial regulatory effectiveness. AI-specific electricity consumption reaches 300–400 TWh by 2030 and 500–700 TWh by 2035, representing 3–5% of global electricity demand. Carbon emissions reach 20–30 $MtCO_2e$ annually by 2030, with modest declines thereafter as renewable energy procurement expands. Water consumption reaches 10–15 billion m^3 annually, creating significant stress in water-scarce regions.

Aggressive Growth Scenario: Under this scenario, AI experiences transformative adoption across all sectors, efficiency improvements lag behind demand growth, and regulatory frameworks fail to constrain externalities. AI-specific electricity consumption exceeds 500 TWh by 2030 and reaches 1,000+ TWh by 2035, comparable to the total electricity consumption of Japan or Germany. Carbon emissions exceed 50 $MtCO_2e$ annually, and water consumption reaches 20–30 billion m^3, creating acute resource conflicts.

The most likely outcome lies between the base case and conservative scenarios, contingent on the effectiveness of sustainable AI strategies documented in Chapter 6 and the regulatory frameworks examined in Chapter 5. Key uncertainties include:

Model Scaling Trajectories: Whether the trend toward ever-larger models continues or shifts toward smaller, specialized models will dramatically influence computational demand. The emergence of efficient small language models (SLMs) and the success of mixture-of-experts architectures could reduce aggregate demand by 50–70%.

Hardware Innovation Pace: The commercialization of neuromorphic computing, photonic processors, and quantum computing could fundamentally alter the energy efficiency landscape, but the timeline for these technologies remains uncertain.

Grid Decarbonization Rate: The carbon intensity of AI computing depends critically on the pace of global electricity grid decarbonization. Accelerated deployment of renewables and nuclear power could reduce AI's carbon footprint even as electricity consumption grows.

Regulatory Stringency: The extent to which jurisdictions implement and enforce efficiency standards, reporting requirements, and carbon pricing will shape industry behavior and investment patterns.

7.2 Research Priorities

Addressing the environmental challenges of AI requires targeted research across multiple disciplines. The following priorities emerge from the analysis presented in this review:

Lifecycle Assessment Methodologies: Current LCA frameworks for AI systems are nascent and inconsistent. Research is needed to develop standardized methodologies for quantifying the full environmental footprint of AI models, including training, inference, hardware manufacturing, and end-of-life disposal. These methodologies should account for geographic variation in grid carbon intensity, cooling water sources, and supply chain emissions.

Rebound Effect Quantification: The magnitude of rebound effects in AI remains poorly understood. Research should develop empirical models that predict how efficiency improvements in AI translate into changes in total resource consumption, accounting for direct, indirect, and economy-wide rebound mechanisms. These models should inform policy design to minimize backfire effects.

Carbon-Aware Algorithm Design: Beyond scheduling and load shifting, research should explore how AI algorithms themselves can be designed to minimize carbon emissions. This includes developing carbon-aware training objectives, dynamic model sizing based on query complexity, and adaptive inference that trades accuracy for energy savings when grid carbon intensity is high.

Sustainable Hardware Materials: Research into alternative materials for AI accelerators reducing dependence on rare earth minerals, cobalt, and other environmentally problematic inputs could address supply chain vulnerabilities while reducing extraction impacts. Biodegradable electronics, recyclable semiconductors, and abundant-elementbased computing represent promising directions.

Water-Free Cooling Technologies: While liquid immersion and direct liquid cooling reduce water consumption, truly water-free cooling systems (thermoelectric cooling, radiative sky cooling, thermosyphon systems) could eliminate the water-energy nexus in data center operations. Research should focus on scaling these technologies to hyperscale deployment.

AI-for-Climate Validation: While the theoretical potential of AI for climate mitigation is substantial, rigorous empirical validation of claimed benefits is lacking. Research should establish standardized protocols for measuring the net emission reductions achieved by AI-enabled systems, accounting for the computational energy consumed by the AI itself and any rebound effects.

Socio-Technical Systems Analysis: AI's environmental impact cannot be understood in isolation from social, economic, and political contexts. interdisciplinary research integrating computer science, environmental science, economics, and policy studies is needed to develop holistic frameworks for sustainable AI governance.

7.3 Policy Recommendations

Based on the evidence synthesized in this review, the following policy recommendations are offered for consideration by governments, international organizations, and industry stakeholders:

Mandatory Environmental Disclosure: All AI service providers and data center operators above a threshold size (e.g., 1 MW power consumption) should be required to disclose energy consumption, water usage, and carbon emissions on a per-service basis. Disclosure should follow standardized methodologies (such as the Green Software Foundation's SCI specification) to enable comparison and benchmarking.

Carbon Pricing for AI Computing: AI computing should be subject to carbon pricing mechanisms that internalize the environmental cost of electricity consumption. This could take the form of carbon taxes on data center electricity, cap-and-trade systems for AI computing capacity, or mandatory carbon offset purchases for AI training runs above a threshold size.

Efficiency Standards and Labels: Minimum efficiency standards should be established for AI data centers, with mandatory labeling of AI services indicating their carbon intensity per unit of output (e.g., grams of CO_2 per 1,000 tokens generated). This would create market incentives for efficiency and enable consumers to make informed choices.

Renewable Energy Requirements: New data centers should be required to procure 100% renewable energy, with a transition to 24/7 carbon-free energy matching over a defined timeline. Existing data centers should face progressively stricter renewable energy requirements.

Water Use Regulations: Data center water consumption should be regulated in water-stressed regions, with requirements for water recycling, use of non-potable sources, and prohibition of freshwater use in areas experiencing severe drought. Water efficiency metrics (liters per kWh) should be reported and benchmarked.

Research and Development Funding: Public funding should support research into sustainable AI technologies, including efficient algorithms, low-carbon hardware, water-free cooling, and circular economy approaches. Funding should prioritize technologies with near-term deployment potential and measurable environmental benefits.

International Coordination: Given the global nature of AI infrastructure and supply chains, international coordination is essential for effective governance. The UNFCCC, OECD, or a new specialized body should develop harmonized standards for AI environmental accounting, facilitate technology transfer for sustainable AI practices, and coordinate research priorities.

Just Transition Frameworks: The transition to sustainable AI must account for distributional impacts on workers, communities, and developing countries. Frameworks should ensure that data center workers receive retraining and support during efficiency transitions, that mineral-rich developing countries benefit from responsible extraction practices, and that the benefits of AI-enabled climate mitigation are equitably distributed.

8. CONCLUSION

The environmental impact of artificial intelligence represents one of the defining challenges of the digital age. This comprehensive review has documented both the substantial environmental costs of AI infrastructure and the transformative potential of AI for climate change mitigation, revealing a complex picture that defies simple characterization.

The evidence is clear that AI's direct environmental footprint is large and growing rapidly. Global data center electricity consumption reached 415-470 TWh in 2024-2025, with AI-specific workloads representing 20-29% of this demand and projected to reach 46% by 2030. The training of frontier models such as GPT-4 consumed approximately 50 GWh of electricity, while inference operations now constituting 80-90% of AI computing energy drive aggregate demand that dwarfs training. Water consumption for data center cooling has escalated dramatically, with hyperscalers reporting 15-25% year-over-year increases and global AI-related water demand projected to reach 4.2-6.6 billion cubic meters by 2027. The manufacturing of AI hardware depends on critical minerals whose extraction carries substantial environmental and social costs, while the rapid obsolescence of AI accelerators generates growing volumes of electronic waste.

Yet the evidence is equally clear that AI offers unprecedented capabilities for addressing climate change. AI-enabled renewable energy forecasting improves accuracy by 15-30%, facilitating greater penetration of wind and solar power. Smart grid management reduces transmission losses and defers fossil fuel infrastructure investments. AI accelerates materials discovery for carbon capture, batteries, and sustainable fuels. Precision agriculture reduces fertilizer overuse and food waste. Advanced climate modeling improves prediction of extreme events and informs adaptation strategies. Transportation optimization cuts fuel consumption across aviation, shipping, and road transport.

The critical question is whether AI's climate benefits exceed its environmental costs a question that cannot be answered with current data. The IEA estimates that AI-enabled energy system optimization could reduce global CO_2 emissions by 1.5-4.0 Gt annually by 2030, while AI's direct emissions are projected at 8-20 $MtCO_2e$ annually. On this basis, AI's mitigation potential appears to far exceed its direct footprint. However, this comparison neglects indirect and higher-order effects, including the Jevons Paradox rebound effects that may cause efficiency improvements to increase total resource consumption, and the embodied emissions of the infrastructure required to deploy AI at scale.

The path toward net-positive environmental outcomes from AI requires concerted action across multiple domains:

Technological Innovation: Continued investment in efficient algorithms, low-carbon hardware, water-free cooling, and circular economy approaches can reduce AI's environmental footprint while maintaining or improving capability. The pipeline of innovations is robust, but commercialization timelines must be accelerated through targeted R&D funding and market incentives.

Regulatory Frameworks: The patchwork of national and regional regulations documented in Chapter 5 must be harmonized and strengthened. Mandatory environmental disclosure, carbon pricing, efficiency standards, renewable energy requirements, and water use regulations can internalize environmental externalities and create market incentives for sustainable practices. International coordination through ISO, UNFCCC, or a new specialized body is essential for effective global governance.

Industry Leadership: Hyperscalers and AI developers must move beyond voluntary commitments to implement measurable, verified sustainability practices. This includes 24/7 carbon-free energy matching, comprehensive lifecycle assessment of AI services, transparent reporting of environmental metrics, and investment in sustainable supply chains. The Sustainable IT Coalition's commitments represent a starting point, but independent verification and binding targets are needed.

Research and Monitoring: The evidence base for AI's environmental impact remains incomplete, particularly regarding rebound effects, embodied emissions, and net mitigation benefits. Sustained investment in interdisciplinary research integrating computer science, environmental science, economics, and policy studies is essential for informed decision-making. Real-time monitoring systems tracking AI energy consumption, water use, and emissions should be established at national and international levels.

Public Engagement: The environmental implications of AI are too significant to be left to technologists and policymakers alone. Public engagement, education, and democratic deliberation are needed to shape the values and priorities that guide AI development. Questions about the appropriate scale of AI deployment, the trade-offs between capability and sustainability, and the distribution of benefits and costs require societal input.

The window for steering AI toward sustainable outcomes is narrow but still open. The decisions made in the next 5-10 years about model development priorities, infrastructure investment, regulatory frameworks, and research funding will determine whether AI becomes a net contributor to climate solutions or an accelerator of environmental degradation. The evidence synthesized in this review suggests that sustainable AI is technically feasible and economically viable, but it requires deliberate choices, sustained commitment, and coordinated action across sectors and jurisdictions.

The stakes could not be higher. Climate change represents an existential threat to human civilization, and the next decade is critical for bending the emissions curve toward net zero. AI could be a powerful tool in this effort or it could become a significant new source of emissions that compounds the challenge. The choice is ours, and the time to act is now.

REFERENCES

[1] International Energy Agency (IEA). (2025). Energy and AI: Special Report. Paris: IEA Publications.

[2] Patterson, D., Gonzalez, J., Le, Q., et al. (2021). Carbon emissions and large neural network training. arXiv:2104.10350.

[3] de Vries, A. (2023). The growing energy footprint of artificial intelligence. Joule, 7(10), 2191–2194.

[4] Luccioni, A. S., Viguier, S., & Ligozat, A. L. (2022). Estimating the carbon footprint of BLOOM, a 176B parameter language model. arXiv:2211.02001.

[5] Wu, C. J., Raghavendra, R., Gupta, U., et al. (2022). Sustainable AI: Environmental implications, challenges and opportunities. Proceedings of Machine Learning and Systems, 4, 795–813.

[6] Dodge, J., Prewitt, T., Des Combes, R. T., et al. (2022). Measuring the carbon intensity of AI in cloud instances. arXiv:2206.05229.

[7] Google LLC. (2025). 2025 Environmental Report. Mountain View, CA: Google.

[8] Microsoft Corporation. (2025). 2025 Environmental Sustainability Report. Redmond, WA: Microsoft.

[9] Meta Platforms, Inc. (2025). Sustainability Report 2024. Menlo Park, CA: Meta.

[10] Amazon.com, Inc. (2025). Amazon Sustainability Report 2024. Seattle, WA: Amazon.

[11] United Nations Regional Information Centre. (2025). AI and Water: The Hidden Cost of Digital Intelligence. Brussels: UNRIC.

[12] Goldman Sachs Research. (2025). AI: Data center power demand to grow 160% by 2030. New York: Goldman Sachs.

[13] Uptime Institute. (2024). Global Data Center Survey 2024. Seattle: Uptime Institute.

[14] Masanet, E., Shehabi, A., Lei, N., et al. (2020). Recalibrating global data center energy-use estimates. Science, 367(6481), 984–986.

[15] Shehabi, A., Smith, S. J., Horner, N., et al. (2018). United States Data Center Energy Usage Report. Berkeley, CA: Lawrence Berkeley National Laboratory.

[16] Mytton, D. (2020). Assessing the suitability of the Greenhouse Gas Protocol for calculation of emissions from public cloud computing workloads. Journal of Cloud Computing, 9(1), 1–12.

[17] Kaack, L. H., Donti, P. L., Strubell, E., et al. (2022). Aligning artificial intelligence with climate change mitigation. Nature Climate Change, 12(6), 518–527.

[18] Rolnick, D., Donti, P. L., Kaack, L. H., et al. (2023). Tackling climate change with machine learning. ACM Computing Surveys, 55(2), 1–96.

[19] Strubell, E., Ganesh, A., & McCallum, A. (2019). Energy and policy considerations for deep learning in NLP. arXiv:1906.02243.

[20] Henderson, P., Hu, J., Romoff, J., et al. (2020). Towards the systematic reporting of the energy and carbon footprints of machine learning. Journal of Machine Learning Research, 21(248), 1–43.

[21] Lacoste, A., Luccioni, A., Schmidt, V., & Dandres, T. (2019). Quantifying the carbon emissions of machine learning. arXiv:1910.09700.

[22] Bender, E. M., Gebru, T., McMillan-Major, A., & Shmitchell, S. (2021). On the dangers of stochastic parrots: Can language models be too big? Proceedings of FAccT, 610–623.

[23] Crawford, K. (2021). Atlas of AI: Power, Politics, and the Planetary Costs of Artificial Intelligence. New Haven: Yale University Press.

[24] Dauvergne, P. (2020). AI in the wild: Sustainability in the age of artificial intelligence. Cambridge, MA: MIT Press.

[25] European Commission. (2024). EU AI Act (Regulation 2024/1689). Brussels: European Union.

[26] European Commission. (2023). Energy Efficiency Directive (recast 2023). Brussels: European Union.

[27] European Commission. (2024). European Sustainability Reporting Standards (ESRS). Brussels: European Union.

[28] US Department of Energy. (2023). Data Center Energy and Water Use Report. Washington, DC: DOE.

[29] White House. (2023). Executive Order on the Safe, Secure, and Trustworthy Development and Use of Artificial Intelligence. Washington, DC: The White House.

[30] National Institute of Standards and Technology (NIST). (2024). AI Risk Management Framework. Gaithersburg, MD: NIST.

[31] NDRC (National Development and Reform Commission). (2021). Guidelines on the Development of Data Centers. Beijing: NDRC.

[32] NDRC. (2022). East Data West Computing Implementation Plan. Beijing: NDRC.

[33] Singapore Economic Development Board. (2022). Data Centre Call for Application. Singapore: EDB.

[34] International Organization for Standardization (ISO). (2024). ISO/IEC 22989: Artificial Intelligence Concepts and Terminology. Geneva: ISO.

[35] Green Software Foundation. (2024). Software Carbon Intensity (SCI) Specification. Portland: Linux Foundation.

[36] IEEE Standards Association. (2025). P2857: Standard for Measuring Energy Efficiency of Machine Learning Workloads. Piscataway, NJ: IEEE.

[37] United Nations. (2024). Global Digital Compact. New York: United Nations.

[38] OECD. (2024). OECD AI Principles 2024 Update. Paris: OECD Publishing.

[39] Anthropic. (2024). AI and Climate Change: Opportunities and Risks. San Francisco: Anthropic.

[40] OpenAI. (2024). GPT-4 Technical Report. arXiv:2303.08774.

[41] Mistral AI. (2025). Life Cycle Assessment of Mistral Large 2. Paris: Mistral AI.

[42] Touvron, H., Martin, L., Stone, K., et al. (2023). Llama 2: Open foundation and fine-tuned chat models. arXiv:2307.09288.

[43] Dubey, A., Jauhri, A., Pandey, A., et al. (2024). The Llama 3 herd of models. arXiv:2407.21783.

[44] Google DeepMind. (2024). GraphCast: Learning skillful medium-range global weather forecasting. Science, 382(6677), 1416–1421.

[45] Vaswani, A., Shazeer, N., Parmar, N., et al. (2017). Attention is all you need. NeurIPS, 30, 5998–6008.

[46] Brown, T. B., Mann, B., Ryder, N., et al. (2020). Language models are few-shot learners. NeurIPS, 33, 1877–1901.

[47] Kaplan, J., McCandlish, S., Henighan, T., et al. (2020). Scaling laws for neural language models. arXiv:2001.08361.

[48] Hoffmann, J., Borgeaud, S., Mensch, A., et al. (2022). Training compute-optimal large language models. arXiv:2203.15556.

[49] Sanh, V., Debut, L., Chaumond, J., & Wolf, T. (2019). DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter. arXiv:1910.01108.

[50] Tan, M., & Le, Q. (2019). EfficientNet: Rethinking model scaling for convolutional neural networks. ICML, 6105–6114.

[51] Fedus, W., Zoph, B., & Shazeer, N. (2022). Switch transformers: Scaling to trillion parameter models with simple and efficient sparsity. Journal of Machine Learning Research, 23(120), 1–39.

[52] NVIDIA Corporation. (2024). NVIDIA H100 Tensor Core GPU Architecture. Santa Clara, CA: NVIDIA.

[53] NVIDIA Corporation. (2025). NVIDIA B200 and GB200 Platforms. Santa Clara, CA: NVIDIA.

[54] Jouppi, N. P., Young, C., Patil, N., et al. (2023). TPU v4: An optically reconfigurable supercomputer for machine learning with hardware support for embeddings. arXiv:2304.01433.

[55] Cerebras Systems. (2024). Cerebras Wafer Scale Engine 3. Sunnyvale, CA: Cerebras.

[56] Intel Corporation. (2023). Intel Loihi 2: A New Generation of Neuromorphic Computing. Santa Clara, CA: Intel.

[57] IBM Research. (2023). IBM NorthPole: A brain-inspired chip for efficient AI inference. arXiv:2310.00130.

[58] Lightmatter. (2024). Passage: Photonic Interconnect for AI. Boston: Lightmatter.

[59] Ayar Labs. (2024). Optical I/O Solutions for AI. Santa Clara, CA: Ayar Labs.

[60] US Geological Survey (USGS). (2025). Mineral Commodity Summaries 2025. Reston, VA: USGS.

[61] IEA. (2025). Critical Minerals Market Review 2025. Paris: IEA Publications.

[62] Forti, V., Baldé, C. P., Kuehr, R., & Bel, G. (2020). The Global E-waste Monitor 2020. Bonn: United Nations University.

[63] Basel Convention. (2019). Technical Guidelines on the Environmentally Sound Management of Waste Consisting of Elemental Mercury and Wastes Containing or Contaminated with Mercury. Geneva: UNEP.

[64] Responsible Minerals Initiative (RMI). (2024). Responsible Minerals Assurance Process. Alexandria, VA: RMI.

[65] World Economic Forum. (2024). AI for Agriculture: Scaling Impact for Food Security. Geneva: WEF.

[66] FAO. (2023). The State of Food and Agriculture 2023. Rome: Food and Agriculture Organization.

[67] John Deere. (2024). See & Spray Technology: Precision Agriculture with AI. Moline, IL: John Deere.

[68] Bayer AG. (2024). Climate FieldView: Digital Farming Platform. Leverkusen: Bayer.

[69] Tesla, Inc. (2024). Autobidder: AI-Powered Energy Trading. Austin, TX: Tesla.

[70] California ISO. (2024). 2024 Annual Report on Market Issues and Performance. Folsom, CA: CAISO.

[71] Duke Energy. (2024). Grid Modernization and AI Implementation. Charlotte, NC: Duke Energy.

[72] National Grid. (2024). Network Innovation Allowance Annual Report. London: National Grid.

[73] OhmConnect. (2024). Demand Response Platform Annual Impact Report. San Francisco: OhmConnect.

[74] US Department of Energy. (2024). Grid Modernization Initiative: AI Applications. Washington, DC: DOE.

[75] Kim, S., Chang, J., & Kim, K. (2023). Deep learning for CO_2 capture materials screening. Nature Communications, 14(1), 2847.

[76] Climeworks AG. (2024). Orca Direct Air Capture Facility: Operational Report. Zurich: Climeworks.

[77] MIT Climate and Energy Consortium. (2024). AI-Accelerated Materials Discovery for Carbon Capture. Cambridge, MA: MIT.

[78] Berkeley Lab. (2024). Machine Learning for Sorbent Materials Development. Berkeley, CA: LBNL.

[79] Petra Nova. (2024). Carbon Capture Facility Performance Report. Houston, TX: Petra Nova.

[80] Google Flood Forecasting Initiative. (2024). Annual Impact Report. Mountain View, CA: Google.

[81] UPS. (2024). ORION Route Optimization: 10 Years of Innovation. Atlanta: UPS.

[82] Surtrac (Rapid Flow Technologies). (2024). Adaptive Traffic Signal Control: Pittsburgh Results. Pittsburgh: Rapid Flow.

[83] Tesla, Inc. (2024). Autopilot Energy Efficiency Analysis. Austin, TX: Tesla.

[84] Google Maps. (2024). Eco-Friendly Routing: Impact Assessment. Mountain View, CA: Google.

[85] Google Flights. (2024). Contrail Avoidance: Pilot Program Results. Mountain View, CA: Google.

[86] IEA. (2024). Transport and AI: Pathways to Decarbonization. Paris: IEA Publications.

[87] Jevons, W. S. (1865). The Coal Question: An Inquiry Concerning the Progress of the Nation, and the Probable Exhaustion of Our Coal-Mines. London: Macmillan.

[88] Khazzoom, J. D. (1980). Economic implications of mandated efficiency in standards for household appliances. Energy Journal, 1(4), 21–40.

[89] Brookes, L. (1990). The greenhouse effect: The fallacies in the energy efficiency solution. Energy Policy, 18(2), 199–201.

[90] Saunders, H. D. (1992). The Khazzoom-Brookes postulate and neoclassical growth. Energy Journal, 13(4), 131–148.

[91] Gillingham, K., Kotchen, M. J., Rapson, D. S., & Wagner, G. (2013). The rebound effect and energy efficiency policy. Review of Environmental Economics and Policy, 7(1), 68–88.

[92] Sorrell, S. (2009). Jevons' Paradox revisited: The evidence for backfire from improved energy efficiency. Energy Policy, 37(4), 1456–1469.

[93] Font Vivanco, D., Kemp, R., & van der Voet, E. (2016). How to deal with the rebound effect? A policy-oriented approach. Energy Policy, 94, 114–125.

[94] Greening, L. A., Greene, D. L., & Difiglio, C. (2000). Energy efficiency and consumption The rebound effect—A survey. Energy Policy, 28(6–7), 389–401.

[95] The Shift Project. (2019). Lean ICT: Towards Digital Sobriety. Paris: The Shift Project.

[96] Freitag, C., Berners-Lee, M., Widdicks, K., et al. (2021). The real climate and transformative impact of ICT: A critique of estimates, trends, and regulations. Patterns, 2(9), 100340.

[97] Coroama, V. C., & Mattern, F. (2019). Digital rebound—Why digitalization will not redeem us the environmental impacts of ICT. Proceedings of EnviroInfo, 1–10.

[98] electricityMap. (2025). Real-Time Carbon Intensity Data. Copenhagen: electricityMap.

[99] WattTime. (2025). Automated Emissions Reduction API. Oakland, CA: WattTime.

[100] EPRI (Electric Power Research Institute). (2024). Carbon-Aware Computing: Potential and Pathways. Palo Alto, CA: EPRI.

[101] Allen Institute for AI. (2024). Green AI: Measuring and Reducing the Carbon Cost of NLP Research. Seattle: AI2.

[102] Schwartz, R., Dodge, J., Smith, N. A., & Etzioni, O. (2020). Green AI. Communications of the ACM, 63(12), 54–63.

[103] Cao, Y. S., Choe, J., & Marshall, A. (2020). A survey on generative adversarial networks for imbalance problems in computer vision tasks. Journal of Big Data, 7(1), 1–36.

[104] World Resources Institute. (2024). State of Climate Action 2024. Washington, DC: WRI.

[105] IPCC. (2023). Climate Change 2023: Synthesis Report. Geneva: IPCC.

[106] UNFCCC. (2024). Nationally Determined Contributions Under the Paris Agreement. Bonn: UNFCCC Secretariat.

[107] Carbon Brief. (2025). Analysis: How AI is Changing Global Energy Demand. London: Carbon Brief.

[108] Nature Climate Change. (2024). Editorial: The Environmental Cost of AI. Nature Climate Change, 14(1), 1–2.

[109] Science. (2024). Policy Forum: Regulating AI's Energy Use. Science, 383(6683), 485–488.

[110] Nature. (2025). Outlook: Sustainable AI. Nature, 635(8037), S1–S8.

APPENDIX A: DATA TABLES

This appendix provides supplementary data tables referenced in the main text, offering additional detail for researchers and policymakers seeking granular information on AI's environmental footprint.

Table A.1: Global Data Center Electricity Consumption by Region (2024)

Region	Data Center Capacity (GW)	Electricity Consumption (TWh)	AI Share (%)	Growth Rate (%/year)
North America	12.5	165	28	12
Europe	8.2	95	22	10
Asia-Pacific	10.8	125	25	15
China	7.5	85	30	18
Middle East & Africa	1.5	15	15	20
Latin America	0.8	10	12	14
Global Total	41.3	495	26	13

Table A.2: AI Model Training Costs and Emissions by Facility Location

Facility Location	Grid Carbon Intensity (gCO_2/kWh)	Training Cost for GPT-4-class Model (tCO_2e)	Renewable Energy Share (%)
Norway (Lefdal)	15	750	98
Sweden (Luleå)	25	1,250	95
France (Paris)	40	2,000	92
Canada (Quebec)	120	6,000	99
UK (London)	200	10,000	55
Germany (Frankfurt)	380	19,000	52
US (Virginia)	400	20,000	45
China (Inner Mongolia)	550	27,500	65
India (Mumbai)	708	35,400	25
South Africa (Cape Town)	900	45,000	15

Table A.3: Projected AI Infrastructure Growth (2025–2035)

Metric	2025	2027	2030	2035
AI Server Installed Base (M units)	4.5	7.2	12.0	22.0
AI Data Center Power (GW)	25	40	70	130

AI Electricity Consumption (TWh)	135	220	400	780
AI Water Consumption (Bm^3)	5.0	7.5	12.0	22.0
AI CO_2 Emissions ($MtCO_2e$)	5	8	15	28
AI E-Waste Generation (kt)	150	280	500	950

APPENDIX B: GLOSSARY OF TERMS

ASIC (Application-Specific Integrated Circuit): A chip designed for a particular use, rather than intended for general-purpose use. AI ASICs such as Google's TPU are optimized for neural network computations.

Carbon-Aware Computing: The practice of scheduling computational workloads to coincide with periods of low grid carbon intensity, reducing emissions without reducing computational output.

CBAM (Carbon Border Adjustment Mechanism): A European Union policy that places a carbon price on imports of certain goods from non-EU countries, equalizing the carbon cost between domestic and imported products.

CCUS (Carbon Capture, Utilization, and Storage): Technologies that capture CO_2 emissions from industrial processes or directly from the atmosphere, then either utilize the captured CO_2 or store it permanently underground.

DLC (Direct Liquid Cooling): A cooling technology that circulates coolant through cold plates attached directly to high-heat components such as CPUs and GPUs, improving heat transfer efficiency.

E-Waste: Electronic waste, or discarded electrical or electronic devices. AI hardware contributes to e-waste through rapid obsolescence cycles.

GPU (Graphics Processing Unit): A specialized electronic circuit designed to rapidly manipulate and alter memory to accelerate the creation of images. GPUs have become the dominant platform for AI training due to their parallel processing capabilities.

Inference: The process of using a trained AI model to make predictions or generate outputs. Inference typically consumes 80–90% of total AI computing energy.

Jevons Paradox: The phenomenon whereby improvements in the efficiency of using a resource lead to increased consumption of that resource rather than decreased consumption.

LLM (Large Language Model): A type of AI model trained on vast amounts of text data to understand and generate human language. Examples include GPT-4, LLaMA, and Claude.

MoE (Mixture of Experts): A neural network architecture that activates only a subset of model parameters for each input, reducing computational cost while maintaining model capacity.

$MtCO_2e$ (Million Tonnes of CO_2 Equivalent): A standardized unit for measuring greenhouse gas emissions, expressing the global warming potential of various gases in terms of equivalent tonnes of carbon dioxide.

PUE (Power Usage Effectiveness): The ratio of total facility energy to IT equipment energy in a data center. A PUE of 1.0 would indicate perfect efficiency with no overhead.

Quantization: The process of reducing the numerical precision of model weights, typically from 32-bit floating point to 8-bit or 4-bit integers, to reduce model size and inference energy.

Rebound Effect: The reduction in expected gains from new technologies that increase the efficiency of resource use, because of behavioral or other systemic responses.

Scope 1/2/3 Emissions: The GHG Protocol classification: Scope 1 = direct emissions from owned sources; Scope 2 = indirect emissions from purchased electricity; Scope 3 = all other indirect emissions in the value chain.

TWh (Terawatt-Hour): A unit of energy equal to one trillion watt-hours, commonly used to measure large-scale electricity consumption. Global electricity consumption is approximately 25,000 TWh annually.

TPU (Tensor Processing Unit): Google's custom-developed AI accelerator chip, designed specifically for neural network computations.

Transformer: A deep learning architecture introduced in 2017 that uses self-attention mechanisms to process sequential data. Transformers underpin most modern large language models.

WEEE Directive: The EU Waste Electrical and Electronic Equipment Directive, which sets collection, recycling, and recovery targets for electrical goods and is part of extended producer responsibility legislation.